RACKWIRE

USER MANUAL

Plan the system. Connect its parts. Catch the conflict.

RackWire plans network, power, and other interconnects across supported devices. Rack layouts are optional. This guide covers the logical map, capacity checks, snapshots, imports, exports, and live telemetry.

01 · QUICK START

Build a useful plan in four moves

1. Select **Add gear**, search the library, and add a device.
2. On the Map, select one port and then the port where the run lands.
3. Choose the cable standard, length, status, and optional label.
4. Open **Issues** to resolve connector, length, power, and capacity conflicts.

> **Safety boundary**
> RackWire is a planning and documentation aid. Confirm cable ratings, electrical loads, PoE classes, equipment requirements, and site conditions before installation.

02 · WORKSPACE

Map for topology. Data for precision.

Map is the spatial planning surface. Drag empty space to pan, use the zoom controls or wheel to zoom, and select **Fit** to frame the plan. Use Move to arrange items, Select for multi-selection, Note for annotations, and Rack for optional rack outlines. A plan can cover a whole site or connect devices that are not in one rack.

Data exposes devices, connections, and issues as editable rows. Filters narrow the current list without changing the plan.

The application chrome can use a light or dark host theme. The planning canvas always remains **Rack Ink #070A0E**. This fixed field preserves the specified contrast and meaning of every cable hue.

03 · DEVICES AND RACKS

Place gear, then describe the load

The Gear library ships with a curated catalog of 244 devices at the reviewed catalog revision spanning the whole environment — not only rack equipment, but routers and switches, servers and single-board computers, workstations, tablets and phones, displays and televisions, appliances, and audio and lighting hardware. Search matches on name, model, vendor, kind, or UPC, so typing a maker (*ubiquiti*, *apple*) or a category (*switch*, *display*) narrows the list as quickly as a model number does. Adding an item creates a plan-owned snapshot of its device definition, so later library changes do not silently rewrite an existing plan.

Select a device to edit its name and planning values. Depending on the device, these include power draw, PoE budget, circuit capacity, a live telemetry key, and port mappings. Capacity badges on the map show planned values until live measurements are available.

Barcode workflow

Select **Scan** to use a camera or enter a code manually. Map a new code to a library device once; RackWire retains that mapping with the plan.

04 · CONNECTIONS

Land each run deliberately

Select a source port and a destination port. RackWire opens the cable editor after the second endpoint is chosen. Set the standard and length, mark the run Planned or Installed, and add a label or callout where useful.

An unconnected endpoint is intentionally bright yellow. It is not an error color and it is never reused for a cable domain.

05 · SIGNAL SYSTEM

Color states what a run carries

Signal colors are fixed brand literals. A host theme may repaint controls and panels, but must never remap these meanings.

Network / data	#22D3EE
Power	#FB923C
USB / serial	#2E6BE6
Other	#74889D
Unconnected end	#FFE04D
Lighting · Studio reserved	#E879F9
Audio · Studio reserved	#A3E635

06 · ANALYSIS

Resolve the warning, then verify the field

RackWire checks connector compatibility, negotiated link limits, cable length, PoE delivery and device budget, AC continuous load at an 80% planning derate, adapter loss, uplink oversubscription, missing power feeds, and double-landed ports.

Issue severity describes the plan model. It does not certify a physical installation. Correct the plan, then validate ratings and measurements against manufacturer and site documentation.

07 · DATA VIEW

Edit and audit at row level

Switch between Devices, Connections, and Issues. Use the filters to focus on a domain or installation state. Connection rows expose endpoints, cable standard, length, labels, status, and live values when present.

CSV export is useful for a punch list or spreadsheet review. CSV import updates supported tabular fields; JSON is the complete RackWire plan interchange format.

08 · NETWORK DISCOVERY

Scan, review, confirm, then place

Choose **Discover devices on the network…** from the main menu when RackWire is served with its optional discovery backend. In the SolariNet dashboard embed, the same action opens SolariNet’s bounded, queued network discovery screen, where an operator can review results and adopt hosts into monitoring. This network discovery path is separate from scanner-ingest: scanner-ingest captures physical inventory, and its reviewed import writes to the shared SoR inventory and barcode registry. RackWire can resolve those barcodes when the SoR model has a RackWire device profile. Configure or update that profile from **Inventory → Components → RackWire device profiles**; verify each port group against the hardware specification before planning. Adopting a discovered host adds it to monitoring and does not create physical inventory. A local `file://` copy retains the planner but cannot call hosted services.

1. Check that the discovery service is available. Its subnet allowlist must include the network you administer; an empty allowlist denies every scan.
2. Start a scan and follow its progress. The service combines available ARP, ping, port, SNMP, and LLDP observations; missing system tools yield less evidence.
3. Review each candidate. Confirm or reject it, edit uncertain make/model/ports, then associate it with an existing library entry or save a validated new model and place it in the plan.

The optional Claude inference path may suggest details for uncertain devices; deterministic evidence and the operator's confirmation remain authoritative. No discovered device is imported automatically. RackWire discovery has its own SQLite database and does not read or write SolariNet's SoR.

> **Hosted service boundary**
>
> The current discovery backend is intended for a single-operator intranet deployment. It requires an authorized subnet, uses a CSRF header for writes, and still needs full multi-user authentication and scan rate limits before public deployment.

09 · OPTIONAL LIVE TELEMETRY

Measurements overlay the plan

Select **Live** and choose an available source. The built-in simulator is suitable for exercising the interface. HTTP polling and WebSocket sources require a compatible telemetry endpoint.

Live values are labeled as live and can show reachability, negotiated rate, throughput, errors, delivered PoE, device load, temperature, and CPU. Missing telemetry remains distinct from a zero measurement.

10 · SAVE AND EXCHANGE

Autosave locally; export intentionally

The browser’s local storage is the standalone store and is updated as the plan changes. Menu → **Export plan (JSON)** creates a portable backup; **Import plan (JSON)** replaces the current working plan with a RackWire export.

In the SolariNet dashboard, RackWire keeps a browser recovery copy and synchronizes the active plan to the shared SoR plan service. Wait for **Shared plan saved** before closing the tab when a save matters. If the service is offline, your local copy remains available and the status tells you that the shared copy did not save. Reopen the dashboard when the connection returns; RackWire will retry an interrupted save from its retained local copy.

Shared plans use a version check so two browsers cannot silently overwrite each other. If another operator saves first, RackWire pauses synchronization and keeps your edits locally. Choose **Save local edits as a separate shared plan…** to preserve both copies, or choose **Load shared plan · archive local edits…** to save a local snapshot before loading the latest shared plan. If an earlier request finishes after you save a separate copy, RackWire reports that the earlier edits also reached the previous plan.

If a save may have reached the server but its reply was lost, an identical retry is accepted safely. If you edit again before resolving that uncertain save, the newer copy can meet a version conflict; reload the shared version or save your local edits as a separate plan to reconcile it. If a shared plan was deleted, RackWire retains the local copy and waits for you to save it explicitly as a new shared plan.

Snapshots can be stored locally or on the shared plan. A shared snapshot is created only after the current edits have synchronized. Restoring a shared snapshot checks that the plan has not changed, waits for any active save, and preserves the current state in Undo. A concurrent edit during restore remains local and requires review before it can synchronize. The standalone `file://` app stays local and does not contact the SoR.

> **Before clearing browser data**
> Export a JSON copy. Browser privacy controls, storage cleanup, or changing origins can make a locally saved plan unavailable.

11 · KEYBOARD

Common shortcuts

KEYS	ACTION
`⌘/Ctrl` + `Z`	Undo
`⌘/Ctrl` + `Shift` + `Z`	Redo
`⌘/Ctrl` + `F` or `/`	Find a device or port
`⌘/Ctrl` + `S`	Export the plan as JSON
`Delete`	Delete the selected item
`Esc`	Cancel the active operation or close a panel

12 · RUNTIME

Works from a file and without a CDN

Open `index.html` directly for standalone use. The application, React runtime, and brand assets are stored in the repository, so core planning works offline. Camera access and live network sources remain subject to browser security and origin rules.

In the SolariNet same-origin iframe, use the dashboard's **Scan inventory barcode** action to open RackWire's barcode lookup, then resolve a scanner-ingested asset against the SoR and retain its asset tag and serial on the placed device. Plan persistence, inventory lookup, library merge, and telemetry use separate SolariNet endpoints when available. Offline planning falls back to local browser storage. The standalone discovery backend remains independent of those endpoints.

UNVERIFIED: live hosted discovery, browser camera permissions, deployed SolariNet iframe/API behavior, and public deployment readiness. Brand Standards v1.1 governs visual usage.

13 · TROUBLESHOOTING

Find the boundary that failed

SYMPTOM	CHECK
Discover cannot connect	The planner still works offline. Discovery requires the hosted RackWire service at the same origin; opening `index.html` from a file does not start that backend.
Scan is denied	Confirm the requested network is in the operator-configured subnet allowlist. An empty allowlist denies all scans. Missing scanner tools can also reduce results without stopping the planner.
Live fields are blank	Confirm a live source is selected and its endpoint is reachable. An omitted field means no measurement was supplied; it is not a zero reading. Use the simulator to check the overlay itself.
A plan seems missing	Check the exact browser origin and profile. Local storage belongs to that origin. Import a prior JSON export if one exists; do not clear browser data before exporting.